

MIAE 221: Materials Science for Engineers

Chapter 7: Dislocations and Strengthening Mechanisms (Expanded & Condensed Guide)

Concordia University · Gina Cody School of Engineering

Textbook:

Materials Science and Engineering: An Introduction

(10th Ed., Callister & Rethwisch)

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(Syllabus Week 7 · Sections 7.1–7.4 · Midterm & Final Scope)

1. 🎯 30-Second Core Intuition & Plain-English Concept

Plastic deformation in metals occurs when line defects (dislocations) slide across close-packed atomic planes.

The Universal Law of Metallurgy: To make a metal stronger, you must **impede dislocation motion**. If dislocations cannot move, the metal cannot permanently deform—its yield strength skyrockets!

There are **Four Universal Strengthening Mechanisms**:

1. **Grain Size Reduction:** Grain boundaries block dislocations.
2. **Solid Solution Strengthening:** Solute atoms create lattice strain fields that pin dislocations.
3. **Strain Hardening (Cold Work):** Tangling dislocations into each other.
4. **Precipitation Hardening:** Hard microscopic particles act as barricades.

2. ⚙️ High-Yield Mathematical Engine & Governing Laws

1. Schmid's Law for Resolved Shear Stress

$$\tau_R = \sigma \cos \phi \cos \lambda$$

- σ : Applied macroscopic tensile stress.
- ϕ : Angle between tensile axis and the **normal to the slip plane**.
- λ : Angle between tensile axis and the **slip direction**.
- $(\cos \phi \cos \lambda)$: **Schmid Factor (m)**. Maximum theoretical value is 0.5 (when $\phi = \lambda = 45^\circ$).
- Yielding initiates in a single crystal when $\tau_R = \tau_{\text{crss}}$ (Critical Resolved Shear Stress):

$$\sigma_y = \frac{\tau_{\text{crss}}}{(\cos \phi \cos \lambda)_{\text{max}}}$$

2. Hall-Petch Relationship (Grain Size Reduction)

$$\sigma_y = \sigma_0 + k_y d^{-1/2}$$

- σ_0 : Friction stress resisting dislocation motion in a single crystal.
- k_y : Strengthening coefficient (material constant).
- d : Average grain diameter. Smaller grain size $d \implies$ higher yield strength σ_y .

Unique Advantage

: Grain size reduction is the **only strengthening mechanism that increases both strength AND toughness simultaneously!**

3. Strain Hardening (Cold Work)

- Flow stress in plastic region: $\sigma_T = K \epsilon_T^n$ (n : strain-hardening exponent).

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

represent the angle between the normal to the slip plane and the applied stress direction, and let λ be the angle between the slip and stress directions, as indicated in Figure 7.7; it can then be shown that for the resolved shear stress τ_R

$$\tau_R = \sigma \cos \phi \cos \lambda \quad (7.2)$$

where σ is the applied stress. In general, $\phi + \lambda \neq 90^\circ$ because it need not be the case that the tensile axis, the slip plane normal, and the slip direction all lie in the same plane.

A metal single crystal has a number of different slip systems that are capable of operating. The resolved shear stress normally differs for each one because the orientation

Figure 7.7 Geometric relationships between the tensile axis, slip plane, and slip direction used in calculating the resolved shear stress for a single crystal.

Figure 7.7: Geometric relationships between tensile axis, slip plane normal (ϕ), and slip direction (λ) in a single crystal.

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hardness and strength (both yield and tensile) are related to the ease with which plastic deformation can be made to occur, by reducing the mobility of dislocations, the mechanical strength may be enhanced—that is, greater mechanical forces are required to initiate plastic deformation. In contrast, the more unconstrained the dislocation motion, the greater is the facility with which a metal may deform, and the softer and weaker it becomes. Virtually all strengthening techniques rely on this simple principle: *Restricting or hindering dislocation motion renders a material harder and stronger.*

The present discussion is confined to strengthening mechanisms for single-phase metals by grain size reduction, solid-solution alloying, and strain hardening. Deformation and strengthening of multiphase alloys are more complicated, involving concepts beyond the scope of the present discussion; Chapter 10 and Section 11.10 treat techniques that are used to strengthen multiphase alloys.

STRENGTHENING BY GRAIN SIZE REDUCTION

The size of the grains, or average grain diameter, in a polycrystalline metal influences the mechanical properties. Adjacent grains normally have different crystallographic orientations and, of course, a common grain boundary, as indicated in Figure 7.14. During plastic deformation, slip or dislocation motion must take place across this common boundary—say, from grain A to grain B in Figure 7.14. The grain boundary acts as a barrier to dislocation motion for two reasons:

Figure 7.14: Dislocation pile-up at a grain boundary. The atomic mismatch between adjacent grains blocks dislocation slip.

In-Depth Visual Breakdown:

- **Schmid's Law Geometry (Figure 7.7):** Demonstrates that tension does not directly cause slip; rather, slip is driven by the **shear component** resolved onto the slip plane along the slip direction. If the tensile axis is perpendicular to the slip plane ($\phi = 0^\circ \implies \lambda = 90^\circ$), the resolved shear stress is zero ($\cos 90^\circ = 0$), and no slip can occur!
- **Hall-Petch Barrier (Figure 7.14):** When dislocations move along slip plane A in Grain 1, they slam into the grain boundary. Because Grain 2 has a different crystallographic orientation, slip plane B is misaligned. Dislocations pile up at the boundary, generating a back-stress that resists further deformation until applied stress is raised substantially.

4. 🏛️ Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**
 - Fitting two data points to the Hall-Petch equation to find σ_0 and k_y , then predicting σ_y for a third grain size.
 - Calculating and reading resulting yield strength, tensile strength, and ductility from standard empirical curves.
 - The three annealing stages: **Recovery** (internal stresses relieve, conductivity restored), **Recrystallization** (new strain-free grains nucleate, ductility restored, strength drops), and **Grain Growth** (grains coarsen to reduce boundary energy).
- **Concordia Exam Traps:**
 - **Schmid's Law Angle Trap:** In 3D space, $\phi + \lambda \neq 90^\circ$! They are independent angles measured between the tensile axis and two separate vectors. Do not assume $\cos \lambda = \sin \phi$.

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem:

A metal with average grain diameter $d_1 = 0.050 \text{ mm}$ has a yield strength of 135 MPa . When grain size is reduced to $d_2 = 0.008 \text{ mm}$, yield strength increases to 260 MPa . Calculate: (a) The Hall-Petch constants σ_0 and k_y , and (b) The expected yield strength when average grain diameter is 0.002 mm .

- **Step 1: Compute $d^{-1/2}$ for Known Grain Sizes**

$$d_1^{-1/2} = (0.050 \text{ mm})^{-1/2} = 4.472 \text{ mm}^{-1/2}$$

$$d_2^{-1/2} = (0.008 \text{ mm})^{-1/2} = 11.180 \text{ mm}^{-1/2}$$

- **Step 2: Set Up Simultaneous Linear Equations**

$$135 = \sigma_0 + k_y(4.472) \quad \text{--- (1)}$$

$$260 = \sigma_0 + k_y(11.180) \quad \text{--- (2)}$$

- **Step 3: Solve for k_y and σ_0**

$$\text{Subtract (1) from (2) : } 125 = k_y(11.180 - 4.472) = 6.708k_y$$

$$k_y = \frac{125}{6.708} = 18.63 \text{ MPa} \cdot \text{mm}^{1/2}$$

$$\sigma_0 = 135 - (18.63)(4.472) = 135 - 83.31 = 51.69 \text{ MPa}$$

- **Step 4: Predict Yield Strength for $d = 0.002 \text{ mm}$**

$$d_3^{-1/2} = (0.002 \text{ mm})^{-1/2} = 22.361 \text{ mm}^{-1/2}$$

$$\sigma_y = 51.69 + (18.63)(22.361) = 51.69 + 416.58 = 468.3 \text{ MPa}$$
